

Pharmacologic Treatment of Hypertension

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High blood pressure (BP) is a major preventable cause of premature death globally.¹⁻³ Accordingly, the treatment of hypertension is one of the most cost-effective interventions in modern medicine.^{4,5} This chapter relies heavily on the 2017 US American Heart Association/American College of Cardiology (AHA/ACC)⁶ and the 2018 European Society of Cardiology/European Society of Hypertension (ESC/ESH) hypertension guidelines,⁷ referred here as the US and European guidelines, respectively. Both guidelines record the importance of lifestyle interventions prior to and in conjunction with pharmacologic treatment of hypertension, noting that they can delay the development of hypertension and can potentiate the efficacy of BP-lowering therapies (see Chapter 36). Nevertheless, most patients with confirmed hypertension will require lifelong drug treatment, invariably with more than one drug (Table 37.1). International hypertension guidelines have recently converged to prioritize simplicity and pragmatism, although some nuances and differences remain.⁸ This chapter summarizes these recommendations for treatment of hypertension.

DEFINING WHO SHOULD RECEIVE PHARMACOLOGIC TREATMENT FOR HYPERTENSION

BP follows a normal distribution within populations; thus *hypertension* is arbitrarily defined by diagnostic thresholds that are subject to change as new evidence from clinical trials emerges. Hypertension is best defined as *that level of blood pressure at which treatment to lower blood pressure results in significant clinical benefit*. The BP at which treatment results in “significant clinical benefit” for any individual will depend on their absolute cardiovascular (CV) risk.¹⁻⁵ This varies because some people are more vulnerable than others to end organ damage at a given BP. Moreover, some patients (e.g., the frail elderly) will be less tolerant of more aggressive BP lowering than others. Consequently, complex differential BP thresholds and targets emerged in previous guidelines, grouping patients into categories defining their threshold BP for therapeutic intervention and optimal BP goals. In some cases, specific drug classes were also given “compelling indications” and “compelling contraindications” for specific groups of patients. Although useful in tailoring therapy, these recommendations were often misinterpreted as indicating that the specific drug was more important than the achieved BP, which is not the case. As guidelines have evolved, the BP threshold for intervention with drug therapy and the recommended treatment targets have been simplified, and the drug treatment algorithms are more pragmatic and focused on logical combinations of therapy. Key principles underpinning current guidance are that (1) most people need more than one drug to achieve currently recommended BP targets, and (2) the magnitude of BP reduction and the achievement of BP control are the most important drivers of benefit.^{4,5}

Blood Pressure Thresholds for Intervention (Office Blood Pressure)

There is unequivocal evidence that drug treatment of a seated “office” BP of 160/100 mm Hg or higher (grade 2 hypertension) reduces the risk for stroke, myocardial infarction (MI), heart failure, and mortality.⁴⁻⁷ Treating pressures of 140/90 mm Hg or higher, especially in higher-risk patients, is similarly beneficial.⁴⁻⁷ Consequently, most international guidelines define hypertension as an office BP of 140/90 mm Hg or higher.⁹ The exception is the 2017 US hypertension guideline,⁶ which defined stage 1 hypertension as a BP of 130/80 mm Hg or higher. Note that the US hypertension guidelines refers to “stages” of hypertension, whereas other guidelines refer to “grades” of hypertension. The US guideline recommends lifestyle advice for such patients and drug treatment to lower BP to less than 130/80 mm Hg if there is coexisting cardiovascular disease (CVD) or an estimated 10-year risk for CVD greater than 10%.⁶ This was a major change, especially setting the diagnostic threshold for “hypertension” at 130/80 mm Hg or higher, resulting in as many as 50% more people being diagnosed as “hypertensive,” with the majority, especially older patients, recommended for drug treatment. Other international guidelines have retained a diagnostic threshold for hypertension of 140/90 mm Hg for adults.⁹ The European guidance in 2018 remained more cautious about the initiation of drug therapy in patients aged 80 years or older, recommending a treatment threshold of 160/100 mm Hg for previously untreated patients.⁷ However, a 140/90 mm Hg threshold is now more widely adopted, especially for the fit and active elderly in this age group.

Table 37.2 contrasts the US guidance and the 2018 European (ESC/ESH) guidelines, which are representative of most other international guidelines. All guidelines identify a category of patients with “high normal BP” (often referred to as borderline hypertension or prehypertension), which highlights people at high risk for progression to hypertension and in whom lifestyle changes can be beneficial.

Blood Pressure Thresholds for Intervention (Ambulatory and Home Blood Pressure Monitoring)

International guidelines favor the use of out-of-office BP measurement to confirm the diagnosis of hypertension, better identify masked hypertension or “white coat” hypertension, and monitor BP control. When the office BP is elevated, the elevation in BP should ideally be confirmed by ambulatory blood pressure monitoring (ABPM) or home BP monitoring.^{6,7,10,11} This is not necessary when BP is markedly elevated in patients requiring urgent treatment (i.e., in the presence of hypertension-mediated clinical complications or organ damage). ABPM or home BP is also recommended to confirm the quality of BP control, especially in higher-risk patients. Moreover, the US guidelines cite evidence suggesting that home BP monitoring may better engage patients in their treatment and lead to better BP control. The shift

toward increased awareness and use of home BP has accelerated during the COVID-19 pandemic as patients became more isolated from their routine clinic visits, and this is likely to be a continuing trend.

Diagnostic thresholds for hypertension vary according to the method of measurement. When ABPM and home BP monitoring are used to classify hypertension, the diagnostic thresholds are lower than office BP because they represent the average of a greater number of

measurements under different conditions. Table 37.3 summarizes the diagnostic thresholds for hypertension according to different methods of measurement (see Chapter 34).

BLOOD PRESSURE TREATMENT GOALS

There is no single BP treatment goal that will be optimal, achievable, and well tolerated by all. The ideal BP treatment goal is likely to be patient specific. Guidelines should therefore be conservative and pragmatic, curbing the zeal of specialists to advocate ever-lower BP goals, and should only make recommendations supported by solid evidence. Until recently, there was international consensus that two BP goals were appropriate: less than 140/90 mm Hg for those with “uncomplicated hypertension” and a lower goal of less than 130/80 mm Hg for those at higher risk (i.e., patients with diabetes, established CVD or cerebrovascular disease, or chronic kidney disease [CKD]). The SPRINT trial challenged this consensus when it showed that more intensive BP lowering (targeting an office systolic BP <120 mm Hg) was more effective at reducing major CV events and mortality than the current, less intensive systolic BP goal of less than 140 mm Hg.¹² The SPRINT results are supported by a more recent and similar study in older patients from China.¹³ Importantly, BP was measured in SPRINT using a very specific methodology: an automated device after 5 minutes of seated rest in a quiet room (either attended or unattended) followed by three oscillometric measurements. This approach to BP measurement undoubtedly results in lower BP readings compared with usual office BP measurements in clinical practice. Indeed, a recent study comparing routine BP measurement using routine electronic health record data from patients also participating in the SPRINT study showed that routine clinic readings were significantly higher than the SPRINT BP recordings, such that the lower attained systolic BP in the SPRINT cohort (121 mm Hg) was equivalent to 128 mm Hg in contemporaneously routinely measured clinic BP. Furthermore, a mean systolic BP of 135 mm Hg in the less intensively SPRINT cohort group was equivalent to 139 mm Hg using routine clinic BP.¹⁴ These findings concur with the view that the office

TABLE 37.1 Development of Therapeutic Strategies for Hypertension

Year	Therapy
Nondrug	
1920s	Strict low-sodium diet
1929	Lumbar sympathectomy
1944	Kempner rice diet
Drug	
1930s	Veratrum alkaloids
1940s	Thiocyanates
1948	Reserpine, phenoxybenzamine
1950	Ganglion blockers
1951	Monoamine oxidase inhibitors
1958	Thiazide diuretics (chlorothiazide)
1960s	Central α_2 -receptor agonists, non-DHP CCBs, β -blockers
1970s	ACE inhibitors, α_1 -receptor blockers
1980s	DHP CCBs
1990s	ARBs
2000s	Renin inhibitors, ARNIs

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor neprilysin inhibitor; DHP CCB, dihydropyridine calcium channel blocker.

TABLE 37.2 Blood Pressure Classification and Definition of Hypertension According to International Guidelines

BP Category	Systolic (mm Hg)		Diastolic (mm Hg)	BP Threshold (mm Hg)
ESC/ESH Classification of Hypertension (2018)				
Optimal	<120	and	<80	Age <80 yr: 140/90
Normal	120–129	and/or	80–84	Age ≥80 yr: 160/90
High-normal	130–139	and/or	85–89	
Grade 1 hypertension	140–159	and/or	90–99	
Grade 2 hypertension	160–179	and/or	100–109	
Grade 3 hypertension	≥180	and/or	≥110	
Isolated systolic hypertension	≥140	and	<90	
US Classification of Hypertension (2017)				
Normal	<120	and	<80	130/80 if existing CV disease or
Elevated	120–129	and	<80	10-yr CVD risk >10%
Hypertension				140/90 in all others
Stage 1	130–139	or	80–89	
Stage 2	≥140	or	≥90	

BP, Blood pressure; CVD, cardiovascular disease; ESC/ESH, European Society of Cardiology/European Society of Hypertension; DBP, diastolic blood pressure; SBP, systolic blood pressure.

Hypertension grades or stages replace the older terminology of mild, moderate, and severe.

TABLE 37.3 Diagnostic Thresholds for Hypertension According to Different Methods of BP Measurement

Measurement	Systolic BP (mm Hg)		Diastolic BP (mm Hg)	
Office or Clinic	140		90	
24-hour	125–130		80	
Day	130–135		85	
Night	120		70	
Home	130–135		85	
Clinic	HBPM	Daytime ABPM	Nighttime ABPM	24-Hour ABPM
120/80	120/80	120/80	100/65	115/75
130/80	130/80	130/80	110/65	125/75
140/90	135/85	135/85	120/70	130/80
160/100	145/90	145/90	140/85	145/90

24-Hour, daytime and *nighttime* refer to ambulatory blood pressure (BP) averages over these periods. *Home* refers to an average of at least 4 days of seated readings at home, usually two readings, twice per day (i.e., an average of ~16 readings). The top panel shows BP thresholds according to the European 2018 Guidelines. The bottom panel shows the corresponding values of systolic BP and diastolic BP for clinic, HBPM, daytime, nighttime, and 24-hour ABPM measurements according to the US 2017 Guidelines.

ABPM, Ambulatory blood pressure monitoring; HBPM, home blood pressure monitoring.

From Williams B, Mancia G, Spiering W, et al. 2018 ESC/ESH Guidelines for the management of arterial hypertension. *Eur Heart J*.

2018;39(33):3021–3104. Erratum *Eur Heart J*. 2019;40(5):475; and Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in

Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2018;71(19):e127–e248. Erratum *J Am Coll Cardiol*. 2018;71(19):2275–2279.

SBP should be treated to less than 140 mm Hg and ideally less than 130 mm Hg, if tolerated. This is consistent with the US guidance, which recommends a BP target of less than 130/80 mm Hg in all treated patients, irrespective of age or comorbidities.⁶

The ESC-ESH 2018 guidance recommended a target range, noting that the first objective should be to treat all patients to an office BP of less than 140/90 mm Hg and that the optimal systolic BP goal would for most treated patients be from less than 140 mm Hg down to 120 mm Hg, also noting that attempting to lower systolic BP below 120 mm Hg in all patients might be beneficial in some but may be hazardous to others.⁷ The guidance stated that the aim should be to reduce systolic BP to 130 mm Hg if possible, and lower if tolerated (which the guidance noted was likely to be more likely in patients <65 years). Thus, in reality, there is little difference between the US and European guidance for most patients with respect to BP treatment targets, except that the European guidance is more nuanced with respect to the likelihood of achieving the lower end of the BP target range in older and especially frailer patients. The diastolic target in all guidelines is less than 80 mm Hg, although in most patients, this will be less difficult to achieve than the systolic targets. The simple takeaway from all of this is to aim to reduce systolic BP to less than 140/90 mm Hg in all patients, aiming for less than 130/80 mm Hg, and ideally lower if tolerated. [Tables 37.4 and 37.5](#) summarize the treatment targets from the US and European guidelines, respectively.

GUIDE TO SELECTION OF ANTIHYPERTENSIVE AGENTS

Key Principles from Clinical Trials

Some important guiding principles with regard to treatment strategies for hypertension are as follows:

1. Effective BP lowering is overwhelmingly important to reduce risk of major CV events and death in people with hypertension. Thus, the first priority in treatment is to control BP.

2. Early studies focused on diastolic BP as the treatment target, but systolic BP is invariably more difficult to control and more closely linked to CV outcomes; thus it should now be the primary but not the sole focus of treatment.
3. Most patients will require more than one drug (i.e., combination therapy) as part of their treatment strategy, especially with the lower BP targets advocated in the most recent guidance.
4. The response to any class of BP-lowering medication is heterogeneous, with important effects of age and ethnicity, but this heterogeneity is diminished by combination therapy.
5. Some trials have suggested that certain comorbidities (e.g., diabetes) or target organ damage (e.g., left ventricular hypertrophy [LVH], CKD) provide compelling indications for inclusion of specific classes of drug therapy in the treatment regimen, usually renin-angiotensin system (RAS) blockade, but this consideration has become less relevant now that recommended combinations typically include RAS blockade.
6. On average, lowering of BP by 20/10 mm Hg in hypertensive patients will reduce the risk for major CV events by half.
7. The reduction in stroke risk and heart failure appears to follow the predicted reduction in risk based on the epidemiologic association between these morbidities and BP.
8. The observed benefits of BP lowering on coronary events are lower than expected based on epidemiologic predictions, which is best addressed by attention to concomitant risk factors, especially statin therapy.
9. The risk reduction associated with BP lowering is continuous across a wide range of BP, with the benefit from treatment greatest in those with the highest absolute CVD risk. This provides the rationale for advocating the use of complementary strategies to reduce CVD risk (e.g., statins for those with established vascular disease, with target organ damage, or at high calculated CVD risk [i.e., ≥10% during 10 years], and antiplatelet therapy for secondary prevention).

TABLE 37.4 Summary of BP Thresholds and Treatment Goals (Office BP) Recommended by the US 2017 Hypertension in Adults Treatment Guidelines

Clinical Condition(s)	Threshold (mm Hg)	Goal (mm Hg)
General		
Clinical CVD or 10-year ASCVD risk $\geq 10\%$	$\geq 130/80$	$<130/80$
No clinical CVD and 10-year ASCVD risk $<10\%$	$\geq 140/90$	$<130/80$
Older persons (≥ 65 years; noninstitutionalized, ambulatory, community-living adults)	≥ 130 systolic	<130 systolic
Specific Comorbidities		
Diabetes mellitus	$\geq 130/80$	$<130/80$
CKD	$\geq 130/80$	$<130/80$
CKD after kidney transplantation	$\geq 130/80$	$<130/80$
Heart failure	$\geq 130/80$	$<130/80$
Stable ischemic heart disease	$\geq 130/80$	$<130/80$
Secondary stroke prevention	$\geq 140/80$	$<130/80$
Secondary stroke prevention (lacunar)	$\geq 130/80$	$<130/80$
Peripheral arterial disease	$\geq 130/80$	$<130/80$

ASCVD, Atherosclerotic cardiovascular disease; BP, blood pressure; CKD, chronic kidney disease; CVD, cardiovascular disease.

TABLE 37.5 Summary of BP Thresholds and Treatment Goals (Office BP) Recommended by the European 2018 Hypertension in Adults Treatment Guidelines

Age Group (Years)	OFFICE SBP TREATMENT TARGET RANGES (mm Hg)					Office DBP Treatment Target Range (mm Hg)
	Hypertension	+ Diabetes	+ CKD	+ CAD	+ Stroke ^a /TIA	
18–65	Target to 130 (or lower if tolerated but not <120)	Target to 130 (or lower if tolerated but not <120)	Target to <140 –130 (or lower if tolerated)	Target to 130 (or lower if tolerated but not <120)	Target to 130 (or lower if tolerated but not <120)	70–79
65–79 ^b	Target to 130–139 (or lower if tolerated)	Target to 130–139 (or lower if tolerated)	Target to 130–139 (or lower if tolerated)	Target to 130–139 (or lower if tolerated)	Target to 130–139 (or lower if tolerated)	70–79
≥ 80 ^b	Target to 130–139 (or lower if tolerated)	Target to 130–139 (or lower if tolerated)	Target to 130–139 (or lower if tolerated)	Target to 130–139 (or lower if tolerated)	Target to 130–139 (or lower if tolerated)	70–79
Office DBP treatment target range (mm Hg)	70–79	70–79	70–79	70–79	70–79	

^aRefers to patients with previous stroke and does not refer to blood pressure targets immediately after acute stroke.

^bTreatment decisions and blood pressure targets may need to be modified in older patients who are frail and independent.

BP, Blood pressure; CAD, coronary artery disease; CKD, chronic kidney disease; DBP, diastolic blood pressure; SBP, systolic blood pressure; TIA, transient ischemic attack.

Selection of Drug Therapy

Modern guidelines recommend that routine drug treatment of hypertension should be based on 5 major classes of drugs; angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), calcium channel blockers (CCBs), diuretics, and β -blockers, with the latter predominantly reserved for specific indications.^{6,7,9} Recent guidelines also advocate wider use of combinations of BP-lowering drugs, even as initial therapy, to achieve optimal BP control. Thus, the previous lists of compelling indications for individual drug classes for patients with specific comorbidities are less relevant than they were previously. The focus

now is on which combinations of drugs are appropriate for each clinical scenario. Nevertheless, it is important to understand the properties and mechanisms of action of individual drug classes that form the basis of treatment for hypertension, as well as their relative risks and benefits (Table 37.6 and Fig. 37.1) and contraindications for the use of specific classes of BP-lowering therapy in specific clinical situations (Table 37.7) and common adverse effects (Table 37.8). These lists are not comprehensive and are subject to change as new evidence emerges, and up-to-date prescribing information should always be consulted. Fig. 37.2 shows the major sites of action of major antihypertensive drug therapies.

TABLE 37.6 Relative Risk and Benefit of Antihypertensive Drug Classes

Outcome	Thiazide Diuretics	Calcium Channel Blockers	β -Blockers	ACEis/ARBs ^a
Unstable angina	0.89	0.88	0.98	0.97
Myocardial infarction	0.78	0.79	0.85	0.81
Diabetes	0.98	0.80	1.13	0.72
Stroke	0.69	0.65	0.85	0.73
Heart failure	0.53	0.73	0.76	0.64
Death	0.91	0.88	0.93	0.90

^aACE inhibitors and ARBs were grouped as a single class for the purposes of this analysis.

Effectiveness of drugs: 1.0 = no benefit/harm, <1.0 = benefit, and >1.0 = potential harmful effect (from a meta-analysis of major blood pressure-lowering trials conducted for the United Kingdom).

ACEis, Angiotensin-converting enzyme inhibitors; ARBs, angiotensin receptor blockers.

Modified from National Institute for Health and Clinical Excellence [NICE] Hypertension Guideline Development Group, 2006. <https://www.nice.org.uk/guidance/cg127>.

Effects of Reduction in Systolic BP Stratified by Class of Antihypertensive

Studies		Intervention		Control		RR (95% CI)
		Events	Participants	Events	Participants	
Major cardiovascular events						
ACE inhibitor	10	5379	31652	9766	50805	1.03 (1.00–1.06)
ARB	8	3647	27140	3779	29331	0.98 (0.93–1.02)
β-blocker	9	2863	25989	2520	27231	1.17 (1.11–1.24)
CCB	21	7857	63693	12808	82904	0.97 (0.94–0.99)
Diuretic	11	5830	38353	6782	42410	0.97 (0.94–1.00)
All-cause mortality						
ACE inhibitor	14	3321	33104	5865	52263	1.01 (0.97–1.05)
ARB	11	2546	29282	2638	31404	0.99 (0.94–1.04)
β-blocker	12	2805	40953	2688	42170	1.06 (1.01–1.12)
CCB	26	5602	76672	8428	95932	0.97 (0.94–1.00)
Diuretic	12	3425	41625	3806	45707	1.02 (0.97–1.06)

0.5

1

2

Class superior
to pooled comparators

Class inferior
to pooled comparators

0.5 1 2
Class superior to pooled comparators Class inferior to pooled comparators

Fig. 37.1 Effects of Reductions in Systolic Blood Pressure (BP) Stratified by Class of BP-Lowering Drug. ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CCB, calcium channel blocker; CI, confidence interval; RR, relative risk. (From Ettehad D, Emdin CA, Kiran A, et al. Blood pressure lowering for prevention of cardiovascular disease and death: a systematic review and meta-analysis. *Lancet*. 2016;387[10022]:957–967.)

Thiazide and Thiazide-like Diuretics

The class of diuretics includes the traditional thiazide diuretics such as hydrochlorothiazide and bendroflumethiazide, as well as thiazide-like diuretics such as chlorthalidone and indapamide. The latter are termed *thiazide-like* because, like the thiazides, they act primarily by inhibiting the Na⁺-Cl⁻ cotransporter in the distal tubule, promoting sodium excretion, which is integral to their antihypertensive effect. However, the thiazide-like diuretics have different structure than the thiazides and differing actions on other aspects of kidney tubular function, such as carbonic anhydrase inhibition in the proximal tubule. Thiazide and thiazide-like diuretics remain an important therapeutic option for the treatment of hypertension. The early changes in salt and water balance they induce are usually accompanied by counteractivation of several vasoconstrictor mechanisms, including the RAS, which may transiently raise peripheral vascular resistance (PVR) and attenuate BP lowering. Subsequently, a gradual reduction in PVR and a new steady

state of reduced total body sodium and BP are established, usually after about 2 months of treatment.

The sustained actions of these diuretics on the kidney make them preferable to loop diuretics for the control of BP. Although loop diuretics are more potent promoters of acute sodium and water loss, their shorter duration of action can result in compensatory sodium retention during the latter part of the dosing interval, thereby reducing their BP-lowering efficacy. Loop diuretics have no place in the routine management of primary hypertension in patients with well-preserved glomerular filtration rate (GFR). However, thiazide and thiazide-like diuretics lose efficacy in patients with GFR less than 30 mL/min/1.73 m². In such patients, loop diuretics are often required for effective BP lowering, especially when there is clinical evidence of sodium and water retention.

The main adverse effects of thiazide and thiazide-like diuretics are metabolic: hypokalemia, hyponatremia (less frequently),

TABLE 37.7 Contraindications to Specific BP-Lowering Therapies

Pharmacologic Therapy	CONTRAINDICATIONS	
	Compelling	Possible
Thiazide diuretics	Gout	Metabolic syndrome Glucose intolerance Pregnancy
β -Blockers	AV block (grade 2 or 3)	Peripheral artery disease Metabolic syndrome Glucose intolerance Athletes, physically active patients COPD Asthma (use cardioselective β -blocker)
Calcium antagonists (dihydropyridines)		Tachyarrhythmias Heart failure
Calcium antagonists (verapamil, diltiazem)	AV block (grade 2 or 3) Heart failure β -Blocker therapy	
ACE inhibitors	Pregnancy Angioneurotic edema Hyperkalemia Bilateral renal artery stenosis	
ARBs	Pregnancy Hyperkalemia Bilateral renal artery stenosis	
Diuretics (antialdosterone)	Hyperkalemia	CKD stages 4 and 5 ^a
Direct renin inhibitors	Pregnancy Hyperkalemia Bilateral renal artery stenosis	

^aUse only low doses and with extreme caution with respect to potassium levels in patients with advanced CKD.

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; AV, atrioventricular; BP, blood pressure; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease.

impaired glucose tolerance, and small increments in blood levels of low-density lipoprotein (LDL) cholesterol and triglycerides. These diuretics also elevate serum uric acid levels and should be avoided in patients predisposed to gout, as well as in those receiving lithium because of a high risk for lithium toxicity. Lithium reabsorption is similar to sodium in the proximal tubule, and thus distal sodium loss caused by thiazide and thiazide-like diuretics can promote proximal reabsorption of sodium and lithium, potentially leading to lithium toxicity. An incidental advantage of thiazide and thiazide-like diuretics may be reduced risk of osteoporosis, especially in women, as a result of calcium retention.

There has been a recent trend to reduce the recommended dose of these diuretics to minimize these metabolic effects. The dose response for BP to thiazide and thiazide-like diuretics is flat (unlike the adverse effect profile); however, some patients respond well to higher doses, which they tolerate. Moreover, when thiazide or thiazide-like diuretics

TABLE 37.8 Common Side Effects Associated With Various Classes of Antihypertensive Drugs

Class	Side Effects
ACE inhibitors	Cough, hyperkalemia
ARBs	Much less frequent hyperkalemia compared with ACE inhibitors
CCBs	
DHP CCBs	Pedal edema, headache
Non-DHP CCBs	Constipation (verapamil), headache (diltiazem)
Diuretics	Frequent urination, hyperglycemia, hyperlipidemia, hyperuricemia, sexual dysfunction
Central α -agonists	Sedation, dry mouth, rebound hypertension, sexual dysfunction
α -Blockers	Pedal edema, orthostatic hypotension, dizziness
β -Blockers	Fatigue, bronchospasm, hyperglycemia, sexual dysfunction
Potassium channel openers	Hypertrichosis (minoxidil); lupus-like reactions, pedal edema (hydralazine)

ACE, Angiotensin-converting enzyme; ARBs, angiotensin receptor blockers; CCBs, calcium channel blockers; DHP, dihydropyridine.

Principal Site of Action of Major Classes of Blood Pressure-Lowering Drugs



Decreased central sympathetic outflow
Centrally acting α_2 -agonist (e.g., clonidine)
Imidazoline receptor agonists (e.g., moxonidine)
Centrally acting α -methyl dopa



Decreased cardiac output β -blockers, diuretics
Increased vagal tone ACE inhibitors, ARBs, DRIs
Decreased heart rate β -blockers, non-DHP CCBs



Vasorelaxation
ACE inhibitors, ARBs, DRIs, CCBs, α -blockers
Thiazide-type diuretics
Direct vasodilators (e.g., hydralazine, minoxidil)
NEP-inhibitor



Natriuresis—all diuretics, CCBs, NEP inhibitor, SGLT2 inhibitors
Renin inhibition—DRIs

Fig. 37.2 Principal Site of Action of Major Classes of Blood Pressure-Lowering Drugs. ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CCB, calcium channel blocker; DRI, direct renin inhibitor; NEP, neprilysin; non-DHP CCB, nondihydropyridine calcium channel blocker; SGLT2, sodium-glucose cotransporter-2.

are combined with drugs that block the RAS, such as ACE inhibitors or ARBs, the dose response is steeper and higher doses (e.g., hydrochlorothiazide 25–50 mg or chlorthalidone 25 mg) may be especially effective in patients with more resistant hypertension. Guidelines favor the use of thiazide-like diuretics, such as chlorthalidone and indapamide, because they are longer acting than traditional thiazides, such as hydrochlorothiazide, and because there are limited outcome data with the latter when used at low doses.

Potassium-Retaining Diuretics

This class of agents includes spironolactone, eplerenone, and amiloride. Finerenone is a recently approved nonsteroidal mineralocorticoid receptor antagonist that has shown benefits in reducing the progression of diabetic kidney disease¹⁵ and CV morbidity and mortality in patients with diabetes.¹⁶ However, there is less data available on the use of finerenone for the treatment of hypertension.

Spironolactone is an aldosterone receptor antagonist that acts in the distal tubule and collecting ducts, decreasing the reabsorption of sodium and water and decreasing the excretion of potassium. The main action of spironolactone is to decrease tubular expression of epithelial sodium channels (ENaC) and kidney outer medullary potassium channels, and thus it has a relatively slow onset and offset of action. Because its main site of action is on sodium and water handling in distal tubule and collecting ducts, spironolactone as monotherapy is a relatively weak diuretic. Nevertheless, it is effective as a BP-lowering agent although rarely used as initial therapy for hypertension. Spironolactone has the advantage over thiazide-like diuretics in that it does not cause hypokalemia or hyperuricemia and it does not impair glucose tolerance. However, spironolactone has antiandrogen activity by binding to the androgen receptor and preventing it from interacting with dihydrotestosterone. Consequently, it can cause nipple tenderness and gynecomastia in some male patients (~6%) that is dose dependent and can limit its use. Another concern with potassium-sparing diuretics is the risk for hyperkalemia in people with substantially reduced GFR (see later discussion).

Eplerenone is more selective for the aldosterone receptor than spironolactone and consequently avoids its antiandrogen effects. There is very limited experience with the use of eplerenone for the routine management of hypertension. Empirically, milligram per milligram, eplerenone is less potent than spironolactone and less effective at lowering BP.

Amiloride is an antagonist of ENaC in the distal convoluted tubules and collecting ducts, decreasing sodium and water reabsorption and impairing potassium excretion. Previously, amiloride was a popular treatment of primary hypertension, when, like spironolactone, it was often used in combination with thiazide-like diuretics. Amiloride is now less frequently used even though it shares the advantage of spironolactone over thiazide-like diuretics of not causing hypokalemia, hyperuricemia, or impaired glucose tolerance. Amiloride is the treatment of choice in patients with Liddle syndrome.

The decline in popularity of potassium-sparing diuretics for the initial treatment of primary hypertension may reflect the increasing use of ACE inhibitors and ARBs for the management of hypertension and the increased risk for hyperkalemia when these are combined with spironolactone or amiloride, especially in patients with kidney impairment. Spironolactone and amiloride are increasingly used as additional diuretic therapy in multidrug strategies for the treatment of resistant hypertension, when these drugs can be very effective (see “Resistant Hypertension” section).¹⁷⁻¹⁹

β -Adrenoceptor Blockers

β -Blockers reduce BP and CV events in patients with hypertension. Most β -blockers, with the exception of those with strong intrinsic sympathomimetic activity, reduce cardiac output by their negative chronotropic and inotropic effects. As with diuretics, short-term hemodynamic responses can be offset by counteractivation of vasoconstrictor mechanisms, which may limit initial BP lowering. Longer-term reduction in arterial pressure occurs because PVR is restored to pretreatment levels. Partial blockade of renin release from the kidney may contribute to the later hemodynamic response.

β -Blockers differ in their duration of action, selectivity for β_1 receptors, lipophilicity, and partial agonist activity. Side effects include

lethargy, muscle aches with exercise, impaired concentration and memory, aggravation of depression and psoriasis, erectile dysfunction, vivid dreams, and exacerbation of symptoms of peripheral vascular disease and Raynaud syndrome. Nonselective β -blockers are contraindicated in asthma patients and can cause impaired glucose tolerance and worsening of dyslipidemia, notably reduced high-density lipoprotein (HDL) cholesterol and raised triglyceride levels. β -Blockers increase the likelihood of new-onset diabetes, particularly in combination with thiazide-type diuretics.^{20,21} Moreover, meta-analyses have consistently suggested that β -blocker-based treatment of hypertension reduces the risk of CV events (especially the risk of stroke) to a lesser extent than treatment with other major drug classes^{4,22,23} (Fig. 37.1). As a consequence, US and European guidelines both state that β -blockers are not preferred as initial therapy for routine treatment of hypertension and should be used only when there is a compelling indication other than BP control (e.g., in patients with angina, chronic heart failure, or for heart rate control, as well as hypertension).^{6,7}

Another exception is in younger women in whom β -blockers are often effective at lowering BP and are safer than ACE inhibitors or ARBs in women anticipating pregnancy. The good BP-lowering efficacy in younger people most likely reflects higher renin levels, and the BP-lowering action of β -blockers, at least in part, relates to suppression of renin release.

Newer Generation of β -Blockers: β_1 Adrenoceptor Blockade Plus Vasodilation

Concerns regarding β -blockers predominantly reflect studies conducted with unselective or, more commonly, β_1 -cardioselective adrenoceptor blockers, especially atenolol. The newer generation of β -blockers retains β_1 -cardioselective adrenoceptor blockade but also has a vasodilator action, either through (1) associated α -adrenoceptor blocking activity (e.g., carvedilol or labetalol) or (2) nebivolol, via associated β_3 -adrenoceptor agonist activity, which dilates via nitric oxide-mediated mechanisms and also mediates favorable effects on glucose and lipid metabolism, offsetting the unfavorable metabolic effects of nonselective or selective β_1 -adrenoceptor blockade. Labetalol has a rapid onset of action, making it a useful intravenous medication for the treatment of hypertensive emergencies. Labetalol is also safe in pregnancy and in patients with coronary disease because it does not increase heart rate. Labetalol should not be used without prior adequate α -blockade in patients with hyperadrenergic states, such as pheochromocytoma, because unopposed, inadequately blocked α -adrenergic activity can increase BP if β -blockade is not complete.

Carvedilol and nebivolol have been shown to improve outcomes in randomized controlled trials (RCTs) in heart failure.²⁴ Despite this, these agents have not been emphasized in most guidelines because of the lack of clinical outcome data from RCTs demonstrating that they improve outcomes when used to treat hypertension.

Calcium Channel Blockers

CCBs reduce BP,^{4,5} are effective antianginal agents, and have no adverse effects on glucose tolerance or lipid parameters. CCBs are especially effective at smoothing BP variability, which is an independent risk factor for stroke. Systematic reviews have shown CCBs (particularly amlodipine) to be the most cost-effective treatment option for hypertension, mainly because they are the most effective agent at preventing stroke.^{4,5,23} BP response to CCBs is largely determined by the magnitude of BP elevation, more so than with other drugs. Thus, patients with higher baseline BP experience greater BP lowering with CCBs than those with only modest elevations of BP. This property also may explain the smoothing effect of CCBs on BP variability.

There are two main groups of CCBs: the dihydropyridines (e.g., amlodipine, nifedipine) and the nondihydropyridines (e.g., diltiazem, verapamil). The dihydropyridine (DHP) CCBs act mainly by inducing relaxation of arterial smooth muscle by blocking L-type calcium channels, thereby inducing a fall in PVR and arterial pressure. Nondihydropyridine (non-DHP) CCBs block calcium channels in cardiac muscle and reduce cardiac output. Verapamil has an additional antiarrhythmic action through its effects on the atrioventricular node. DHP and non-DHP CCBs have occasionally been combined, but no robust data are available on the BP-lowering efficacy or clinical outcomes of this approach, and it is not a recommended combination therapy in international guidelines.

Earlier formulations of some DHP CCBs, such as capsular nifedipine, had a rapid onset and a short duration of action, with unpredictable effects on BP. These responses were often accompanied by reflex sympathetic stimulation and tachycardia. These shorter-acting oral preparations of CCBs have no place in the routine management of hypertension. Longer-acting formulations of DHP CCBs produce more sustained and predictable responses.

Side effects of DHP CCBs include dose-dependent peripheral edema, which is caused by transudation of fluid from the vascular compartments into the dependent tissues as a result of precapillary arteriolar dilation. This edema does not respond to diuretic therapy but is alleviated by limb elevation or coadministration of an ACE inhibitor or ARB (because of their effects on venous capacitance). Gum hypertrophy can occur with DHP CCBs but is rarely seen with non-DHP CCBs. Non-DHP CCBs cause less peripheral edema but are negatively inotropic and negatively chronotropic and should therefore be avoided in patients with compromised left ventricular function and in combination with β -blockers. Verapamil use is commonly accompanied by constipation.

Renin-Angiotensin System Blockers

Inhibition of RAS is predictably effective at lowering BP by inhibiting the various central and peripheral pressor effects of angiotensin II (Ang II). Blockade of RAS also may lower BP by other mechanisms involving improvements in endothelial function, vagal tone, and baroreceptor function and through inhibition of the kidney tubular reabsorption of sodium. In addition, RAS inhibition has been popularized by clinical trial evidence showing reduced morbidity and mortality in patients with heart failure, delayed progression of kidney disease, and reduced CV events in those at high CV risk.^{6,7,9} Two major classes of drugs directly target the RAS, ACE inhibitors, and ARBs. Direct renin inhibitors were also developed for this purpose but have declined in use and are not recommended for routine treatment of hypertension in international guidelines because of a lack of evidence that they improve CV outcomes.

Angiotensin-converting enzyme inhibitors. ACE inhibitors block the conversion of Ang I to Ang II by inhibiting ACE. The resulting reduction in levels of Ang II leads to vasodilation and a fall in BP. Ang II has many additional actions that are potentially harmful to the CV system and has been implicated in the pathogenesis of structural changes in the heart, blood vessels, and kidneys in hypertension. Sharp falls in BP after the introduction of ACE inhibitors may occur when the RAS is activated (e.g., in patients who are dehydrated, in heart failure, or have accelerated hypertension). This is rarely a problem when therapy is initiated in uncomplicated hypertensive patients.

Side effects of ACE inhibitors include the development of a persistent dry cough in about 20% of users. This is more common in women and in people from East Asia and the Pacific Rim. The cough disappears only after discontinuation of the drug. Another rare but important complication is angioedema, which occurs in about 1% and

is much more common in the Black population (~4%). ACE inhibitors should be avoided in women of childbearing potential because of the danger of fetal malformation, especially from exposure in the first trimester, when women may be unaware they are pregnant. ACE inhibitors should not be used in patients with significant bilateral renal artery disease because they may precipitate deterioration in kidney function and kidney failure. Careful monitoring of kidney function and serum potassium concentration is also required in patients with more advanced kidney impairment of any cause, because of the risk for hyperkalemia.

Angiotensin receptor blockers. ARBs are highly selective inhibitors of the Ang II type 1 receptor (AT_1). As with ACE inhibitors, ARBs inhibit the actions of Ang II on the CV system and kidney. ARBs reduce BP as effectively as ACE inhibitors and generally have a longer duration of action. ACE inhibitors and ARBs appear to be equally effective in reducing albuminuria and preserving GFR²⁴ and have similar efficacy for preventing major CV events in patients with established CVD.^{4,25} Because of their selectivity and specificity for the AT_1 receptor, the ARBs are well tolerated, with a placebo-like adverse effect profile. Cough and angioedema are much less likely to occur with ARBs than with ACE inhibitors, and most guidelines recommend switching patients to an ARB when an ACE-induced cough occurs. Cautions and contraindications are similar to those outlined for ACE inhibitors.

α -Adrenergic Blockers

The original members of the α -adrenergic blocking class (e.g., prazosin) were short-acting drugs that blocked the activation of α_1 -adrenoceptors in the vasculature, leading to vasodilation. Initially, the recommended dosage was too high, and postural hypotension and syncope were frequent. The use of lower doses and the development of longer-acting agents (e.g., doxazosin) have largely overcome this problem. Blockade of sphincteric receptors improves symptoms in men with benign prostatic hypertrophy but can worsen stress incontinence in women. Uniquely among antihypertensive drugs, the α_1 -antagonists produce modest favorable changes in plasma lipids, with a reduction in total and LDL cholesterol and triglyceride levels and an increase in HDL cholesterol. α_1 -Antagonists are not recommended for routine treatment of hypertension and are largely reserved for their specific indications (e.g., prostatic outflow obstruction) or as “add-on” treatment for resistant hypertension when alternative treatments are not tolerated.

Centrally Acting Sympatholytic Drugs

Some of the earliest drugs developed to treat hypertension targeted the activation of the sympathetic nervous system (SNS) at various levels, including the CV regulatory nuclei in the brainstem, the peripheral autonomic ganglia, and the postganglionic sympathetic neuron. Few of these agents are used today because side effects are common.

Methyldopa reduces sympathetic outflow from the brainstem and frequently causes sedation, impaired psychomotor performance, dry mouth, and erectile dysfunction. Its unfavorable effect on quality of life resulted in methyldopa being gradually replaced by more effective drugs, although it is still extensively used in the management of hypertension of pregnancy, which is now its main indication.

Clonidine is now rarely used because of its short duration of action and risk for withdrawal syndrome, which occurs when sudden discontinuation results in a rebound rise in catecholamines with features that may resemble those of pheochromocytoma, such as severe hypertension, tachycardia, and sweating. This is exacerbated when patients are also receiving nonselective β -blockers such as propranolol. The syndrome is treated by readministration of the drug and then gradual

discontinuation or the intravenous infusion of labetalol in an emergency. Clonidine is still used occasionally and can be effective in some patients with resistant hypertension. Extended-release formulations of clonidine are now available.

A newer centrally acting agent, *moxonidine*, is an imidazoline receptor agonist that reduces sympathetic outflow and BP. It has a lower incidence of side effects and is better tolerated than other centrally acting agents. Moxonidine has no clinical trial evidence to support its use as a preferred first-line agent but is used empirically in some patients intolerant of other agents or in resistant hypertension.

Direct Vasodilators

Hydralazine is no longer recommended as a first-line agent for hypertension management. The main disadvantages of hydralazine are sympathetic activation and the development of a lupus-like syndrome, particularly in patients with the slow acetylator genotype. Also, multiple daily dosing is required. It is still very occasionally used in severe hypertension and hypertension associated with pregnancy.

Minoxidil is a potent vasodilator, and its use is largely confined to occasional use in specialist centers for the treatment of severe and resistant hypertension. Its side effect profile includes stimulation of body hair growth; tachycardia and severe fluid retention reflect its potent vasodilator action and concomitant reflex SNS activation. For this reason, minoxidil is usually combined with a potent loop diuretic and a β -blocker as part of a triple-therapy approach to severe hypertension. Long-term use can be associated with insidious development of peritoneal and pericardial effusions (especially in patients with impaired kidney function), which usually respond to treatment withdrawal.

Emerging Drug Therapies for Hypertension

ARB/Neprilysin Inhibition

Treatment with an angiotensin receptor neprilysin inhibitor (ARNI; e.g., sacubitril) has become well established in heart failure and improves patient outcomes compared with prior standard of care.²⁶ The ARNI effectively lowers BP over 24 hours, especially nocturnal BP, and is more effective than ARB alone.^{27,28} The BP lowering effect relates to the mechanism of action whereby neprilysin (NEP) inhibition blocks the breakdown and potentiates the actions of natriuretic peptides, producing a diuretic and vasodilator action that is augmented by concomitant RAS blockade with the ARB component. Although the ARNI has not been widely promoted to treat hypertension, there is ample data showing its efficacy at BP lowering, and it has the potential to be a highly effective and well-tolerated treatment, either alone or in combination with either a CCB and/or thiazide diuretic.

Sodium-Glucose Cotransporter-2 Inhibitors

These drugs (e.g., empagliflozin and dapagliflozin) were developed as treatments to lower blood glucose in patients with diabetes by inhibiting the active proximal convoluted tubular reabsorption of glucose via inhibition of sodium-glucose cotransporter-2 (SGLT2). Thus, SGLT2-inhibition also inhibits the proximal tubular reabsorption of sodium, promoting natriuresis, particularly if additional diuretics are present. This inevitably results in significant reductions in BP that are rarely commented upon but have been clearly documented.^{29,30} It is likely that the natriuresis and BP-lowering effects of SGLT2 inhibition have contributed to improved CV outcomes and kidney outcomes in patients with diabetes, and heart failure outcomes in patients with and without diabetes.³⁰⁻³² In the context of difficult-to-control hypertension, especially in patients with diabetes, SGLT2 inhibition may complement and enhance the BP-lowering effect of other agents, even though not formally recommended for this purpose.

An intriguing aspect of treatment with drugs with diuretic actions that promote natriuresis is that they seem to be particularly effective at reducing nocturnal BP.²⁸ This is often not appreciated as nocturnal BP is rarely recorded unless ABPM is performed. However, this may be important because nocturnal BP is the most powerful predictor of adverse outcomes and death in patients with hypertension and is more likely to be elevated in high-risk patients with diabetes and/or CKD, perhaps because of their predisposition to sodium retention.

Hypertension Treatment Strategies

Despite decades of guidelines for the treatment of hypertension, the control of BP has remained lamentably poor, with typically fewer than 50% of treated patients achieving optimal BP control.³³ This relates to a number of factors that have been inadequately appreciated: (1) the complexity of guidance and overemphasis on specific indications for individual drug therapies versus the overriding importance of getting BP controlled; (2) treatment inertia (i.e., the slow up-titration of therapy and/or failure to escalate treatment by combining drugs to achieve BP control); and (3) poor recognition of the fact that poor adherence to therapy, especially multipill therapy, was a major factor in poor BP control; that is, the BP was not so much resistant to treatment as the patient was resistant to taking multiple drugs.

Recent iterations of the US and European guidance have aligned and dramatically shifted their approach by advocating the following:

1. Similar preferred drug classes for the routine treatment of hypertension, ACE inhibition or ARBs, CCBs, thiazide/thiazide-like diuretics (the US guidance specifically prefers chlorthalidone).
2. Simple and pragmatic treatment algorithms based on treatment with RAS blockade (ACE inhibition or ARB) as the foundation of therapy for most patients, in combination with a CCB and/or a thiazide/thiazide-like diuretic, or all three when required.
3. Initial therapy with two drugs, that is, RAS blockade plus a CCB or thiazide/thiazide-like diuretic for most patients when BP is 140/90 mm Hg or higher and certainly when 20/10 mm Hg or higher above the intended target—the exceptions being the frail elderly where monotherapy may be preferred, or patients with BP just above 140/90 mm Hg, when monotherapy may be sufficient. This recommendation effectively normalizes initiating therapy with two drugs for most people with hypertension.
4. Single-pill combination therapy for dual- and triple-drug therapy, when available at reasonable cost, to reduce the pill burden and improve patient adherence. The European guideline specifically advocates a “single-pill strategy” to treat hypertension.⁷
5. A three-drug combination (i.e., RAS blockade plus CCB plus thiazide/thiazide-like diuretic when the two-drug combination is insufficient).
6. Further diuretic therapy (usually spironolactone unless contraindicated) if further treatment is required (i.e., in resistant hypertension; see “Resistant Hypertension” section).
7. Use β -blockers when there is a guideline-directed indication for their use (e.g., in patients with symptomatic angina, heart failure with reduced ejection fraction, or in patients with atrial fibrillation to facilitate rate control) (Fig. 37.3). The US guidance does not contain a formal treatment algorithm, but the guidance in the US and Europe is very consistent on the treatment strategy⁸ (Table 37.9).

Initial Therapy With a Two-Drug Combination

The lower BP targets recommended in current guidelines makes two-drug combinations almost inevitable for the majority of patients as initial therapy, with the preferred combinations RAS blockade plus diuretic or RAS blockade plus CCB. This will often be a “low-dose” two-drug combination therapy. The recommendation of a two-drug

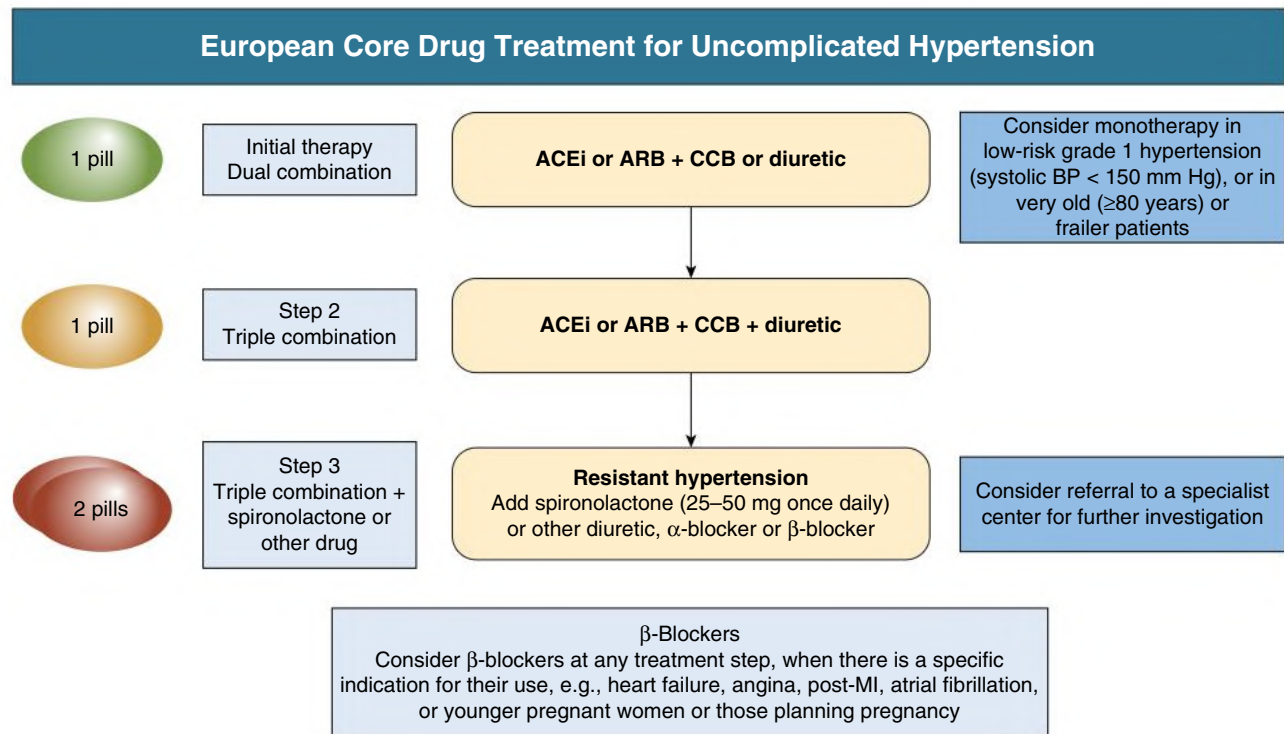


Fig. 37.3 European Core Drug Treatment Algorithm for Uncomplicated Hypertension. This algorithm is also considered appropriate for patients with hypertension-mediated organ damage, cerebrovascular disease, diabetes, or peripheral vascular disease. ACEi, Angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BP, blood pressure; CCB, calcium channel blocker; MI, myocardial infarction. (From Williams B, Mancia G, Spiering W, et al. 2018 Practice guidelines for the management of arterial hypertension of the European Society of Cardiology and the European Society of Hypertension. *Eur Heart J*. 2018;39[33]:3021-3104. Erratum *Eur Heart J*. 2019;40[5]:475.)

TABLE 37.9 US Guidance for Approach to Drug Therapy for Hypertension

Grade	LOE	Recommendation
Recommendation for Choice of Initial Medication		
I	A	For initiation of antihypertensive drug therapy, first-line agents include thiazide diuretics, CCBs, and ACE inhibitors or ARBs.
Recommendation for Choice of Initial Monotherapy Versus Initial Combination Drug Therapy		
I	C	Initiation of antihypertensive drug therapy with two first-line agents of different classes, either as separate agents or in a fixed-dose combination, recommended in adults with stage 2 hypertension and an average BP >20/10 mm Hg above their BP target.
Ila	C	Initiation of antihypertensive drug therapy with a single antihypertensive drug reasonable in adults with stage 1 hypertension and BP goal <130/80 mm Hg with dosage titration and sequential addition of other agents to achieve BP target.

ACE, Angiotensin-converting enzyme; ARBs, angiotensin receptor blockers; LOE, level of evidence.

From Whelton PK, Carey RM, Aronow WS, et al. ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2018;71(19):e127–e248. Erratum *J Am Coll Cardiol*. 2018;71(19):2275–2279.

combination is in part driven by concern that the upward titration of treatment in people at high risk may be too slow and leave them at risk for too long. A two-drug combination is also logical because the response to a single drug is often limited by counteractivation of pressor systems. This also explains why many BP-lowering drugs in monotherapy have a relatively flat dose-response curve. For example, sodium and water loss caused by diuretics or vasodilation with CCBs will activate the renin-angiotensin-aldosterone system, which limits the BP lowering. Thus, a two-drug combination as initial therapy (1) will produce much greater BP lowering than even the most effective monotherapy; indeed, studies have shown this approach is twice as effective as monotherapy at achieving BP control within 3 to 4 months³⁴; (2) will reduce heterogeneity in the BP-lowering response, ensuring that all patients get a response because more than one mechanism is being targeted by a two-drug approach; and (3) will cause the dose-response to be steeper to upward titration of either component because counterregulatory mechanisms are blocked.

The main concern has been adverse events related to potentially large initial BP falls in treatment-naïve patients, but in practice this has not been a major problem, even in older patients, and guidelines already highlight the one group where there may be a need for a more cautious approach to lowering BP (i.e., frail older patients). Ironically, adverse effects may actually be minimized rather than enhanced by the recommended combinations of therapy (i.e., less CCB-mediated peripheral edema) or diuretic-induced hypokalemia when either is combined with RAS blockade.

Further advantages of combination therapy. Now that the major classes of antihypertensive therapy recommended by guidelines are available as generic agents, the costs have reduced substantially, and the same is true for single-pill combinations of two drugs as well

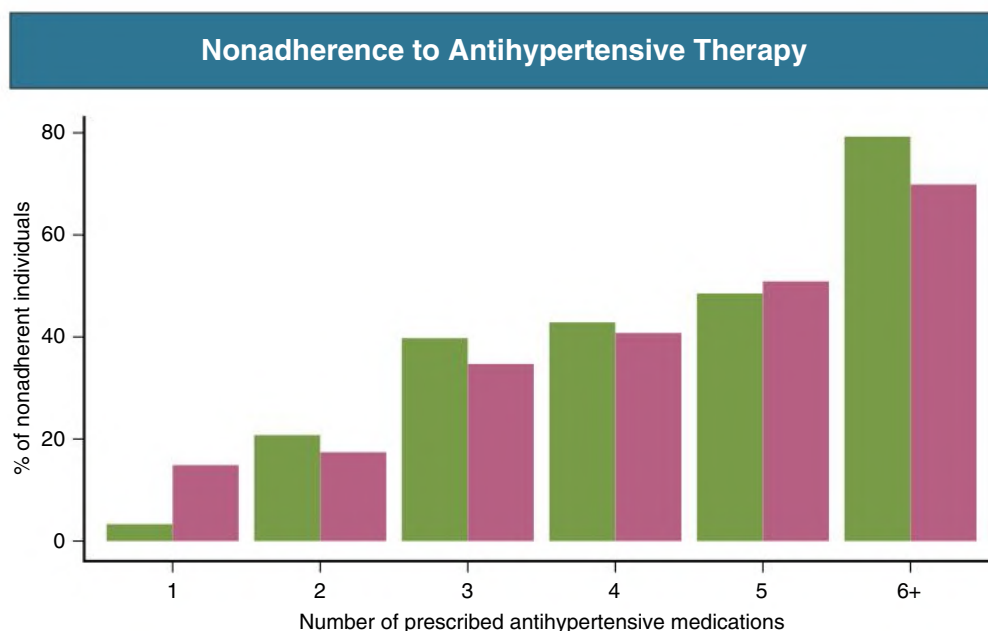


Fig. 37.4 Nonadherence to Antihypertensive Therapy According to Number of Antihypertensive Medications, as Detected by Urine Drug Testing. Green bars denote data from the United Kingdom; red bars denote data from the Czech Republic. (Modified from Gupta P, Patel P, Štrauch B. Risk factors for nonadherence to antihypertensive treatment. *Hypertension*. 2017;69[6]:1113–1120.)

as two drugs. This makes guideline implementation more feasible globally, especially for single-pill combinations, because the number of pills used to treat hypertension is inversely related to patient adherence³⁵ (Fig. 37.4). If a patient's BP is not controlled on three or more drugs, then poor adherence to at least some of the drugs should be considered and excluded as a likely cause of the poor treatment response. The converse is also true; that is, if the patient receives two or three drugs as a single-pill combination, then they are more likely to be adherent to their therapy and are more likely to achieve their BP treatment target.

Registry data shows that patients starting treatment with a two-drug combination are more likely to achieve BP control faster and more likely to escalate to a three-drug combination if needed compared with patients starting treatment with monotherapy and a traditional stepped care approach, many of whom remain on monotherapy despite inadequate BP control. This indicates that the two-drug initial therapy approach, especially with a single-pill combination, is effective at overcoming clinical inertia^{36,37} (Fig. 37.5).

Finally, whereas no RCTs have compared initial combination therapy versus a standard stepped care approach, large-scale registry studies have suggested that initial combination therapy for hypertension is associated with improved CV outcomes compared with a standard initial monotherapy, stepped care approach, most likely due to faster and more effective BP control with the combination therapy³⁸ (Fig. 37.6).

Triple or quadruple low-dose combination therapy as initial therapy for hypertension. Enthusiasm for low-dose combinations of therapy involving multiple drugs is growing, fueled by the philosophy that targeting multiple pressor mechanisms simultaneously produces more effective BP lowering, and that most BP lowering and fewer side effects occur when drugs are used in low dose. Indeed, studies thus far have found that a pill containing four agents (quad pill) in low dose is more effective at lowering BP than standard dose monotherapy,³⁹ but what has not been studied is whether the quad pill would be more effective than low-dose dual therapy, which is now becoming the standard of care and is also much more effective than standard dose monotherapy. Thus, low-dose combination of three or four drugs is considered experimental and not currently recommended.

Combining Renin-Angiotensin System Blockade Is Not Recommended

The view that RAS blockade may help prevent or regress hypertension-mediated structural and functional damage led to the use of dual RAS blockers. However, data from numerous trials has demonstrated that dual RAS blockade was no more effective than RAS blocker monotherapy at preventing major CV events in high-risk populations, including those with diabetes,^{6,7} but increased the risk for adverse events, especially kidney impairment. These findings have prompted all international guidelines to state that dual RAS blockade should not be used to treat hypertension.

Resistant Hypertension

Standardizing the routine treatment of hypertension has also helped refine the definition of resistant hypertension (RH). RH is defined as a BP that remains above target despite the concurrent use of three guideline recommended antihypertensive agents, usually a RAS blocker plus CCB plus thiazide/thiazide-like diuretic in optimal or maximum tolerated doses. The US guidance also includes patients with controlled BP when four different drugs have been required to achieve BP control. The definitions emphasize that causes of “pseudoresistant” or “apparent resistant” hypertension should be excluded. These include (1) ensuring proper measurement of office BP with an appropriately sized cuff (note that a cuff that is too small for the arm can lead to overestimation of BP); (2) the use of ABPM or home BP to exclude a significant white coat effect, which elevates BP only in the doctors' office setting; and (3) exclusion of nonadherence to therapy. Finally, secondary causes of hypertension, such as primary aldosteronism and renovascular disease, should be excluded. Other causes of resistant hypertension also should be considered (Table 37.10). Once this screen is complete, analysis of recruitment into clinical trials of patients with resistant hypertension has consistently shown that as many as 9 out of 10 patients originally considered to have resistant hypertension are excluded, principally because of suboptimal therapy, controlled BP on ABPM or home measurements, or nonadherence to their prescribed therapy. Most patients with resistant hypertension are older, obese, and/or have evidence of target organ damage.^{40–42}

Multidrug Therapy Categorized by Start With Monotherapy or Two-Drug Combination Therapy

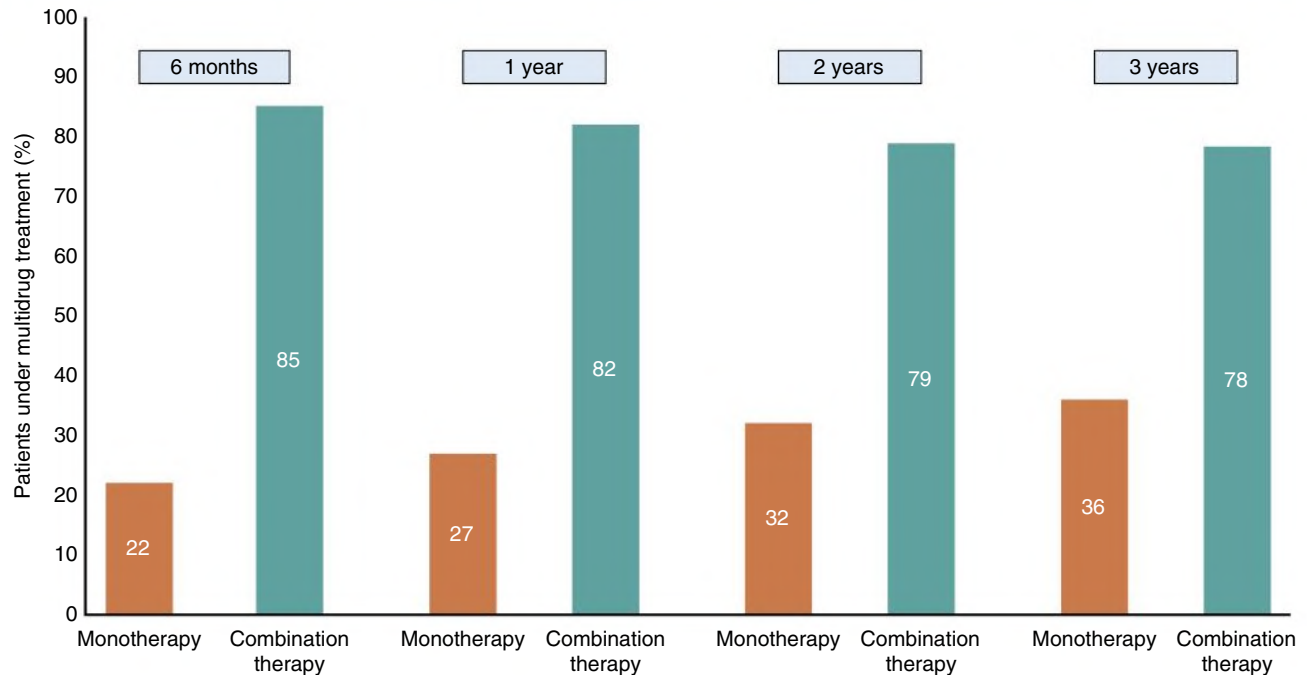


Fig. 37.5 Percent of patients receiving multidrug combination therapy according to whether they started treatment with monotherapy or two-drug combination therapy. (From Rea F, Corrao G, Merlino L, Mancia G. Initial antihypertensive treatment strategies and therapeutic inertia. *Hypertension*. 2018;72[4]:846–853.)

Major Outcomes of Monotherapy or Two-Drug Combination Therapy

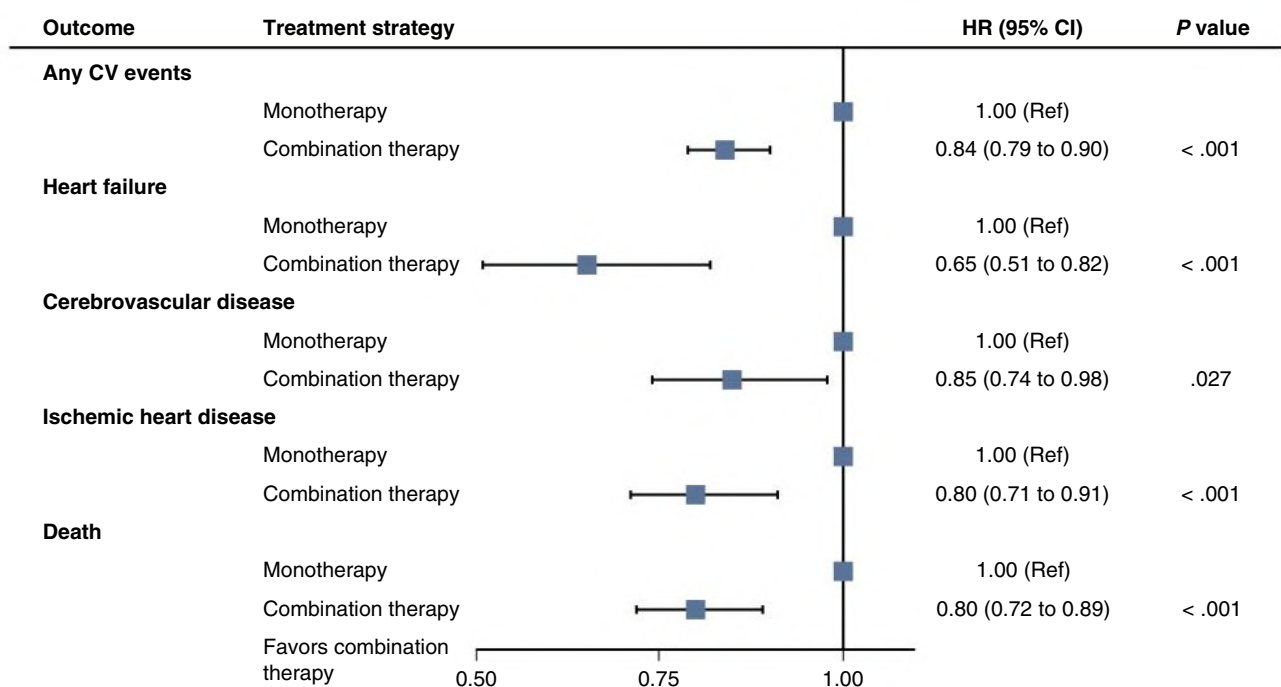


Fig. 37.6 Major outcomes in patients initially treated for hypertension with monotherapy or two-drug combination therapy. CI, Confidence interval; CV, cardiovascular; HR, hazard ratio. (From Rea F, Corrao G, Merlino L, Mancia G. Early cardiovascular protection by initial 2-drug fixed-dose combination treatment vs. monotherapy in hypertension. *Eur Heart J*. 2018;39[40]:3654–3661.)

TABLE 37.10 Considerations in the Patient With Resistant Hypertension

Patient Factors	Secondary Causes of Resistant Hypertension	Concomitant Medications That May Raise BP	Causes of “Pseudoresistant Hypertension”
Demographics	Common causes	Prescription	Poor patient adherence to medications
Older age, especially >75 y	Primary hyperaldosteronism (Conn adenoma)	Oral contraceptives	Check BP response to directly observed medication
Obesity	Atherosclerotic renovascular disease	NSAIDs	Errors in BP Measurement
Women > men	Sleep apnea	Sympathomimetics (e.g., decongestants in cold remedies)	Including BP cuff too small for arm circumference
More common in Blacks	CKD	Cyclosporine	White coat hypertension
Excess dietary sodium	Uncommon Causes	Erythropoietin	Check BP with ABPM or home BP measurements
High baseline BP and chronicity of uncontrolled hypertension	Pheochromocytoma	Corticosteroids (e.g., prednisone, hydrocortisone)	
Concomitant Disease	Aortic coarctation	Nonprescription	
Target organ damage: LVH or CKD	Cushing disease	Drug abuse (e.g., cocaine, amphetamines)	
Diabetes	Hyperparathyroidism	Excess licorice ingestion	
Atherosclerotic vascular disease		Herbal remedies (e.g., ephedra, also known as ma huang)	
Aortic stiffening			

ABPM, Ambulatory blood pressure monitoring; BP, blood pressure; CKD, chronic kidney disease; LVH, left ventricular hypertrophy; NSAIDs, nonsteroidal antiinflammatory drugs.

Modified from Williams B. Resistant hypertension: an unmet treatment need. *Lancet*. 2009;374:1396–1398.

The PATHWAY-2 study of resistant hypertension showed resistant hypertension is most commonly a sodium-retaining state with an inappropriately suppressed plasma renin level, despite background treatment with diuretic therapy and other drugs (i.e., CCB and RAS blockade) that would be expected to raise plasma renin.¹⁹ This has two important therapeutic implications: (1) there should be increased emphasis on dietary sodium restriction, and (2) treatment with further diuretic therapy is likely to be the most effective additional therapy as the fourth-line drug treatment for RH. This was tested and most convincingly proven in the PATHWAY-2 study, which showed that low-dose spironolactone (25 mg daily, titrated to 50 mg daily) was the most effective drug at lowering BP compared with placebo, a β -blocker (bisoprolol), or α -blocker (doxazosin).¹⁷ This emphasizes a key point: although spironolactone is a relatively weak diuretic as a sole agent, when combined with other diuretics acting at a different site in the nephron (as in the treatment of resistant hypertension on a background of thiazide/thiazide-like diuretic), this multisegment nephron blockade can be highly effective at offloading sodium in sodium-resistant states. It is also conceivable that spironolactone was particularly effective because some of these patients may have undetected primary aldosteronism.

However, the PATHWAY-2 study was conducted in patients with normal GFR, and careful monitoring for hyperkalemia is required when using spironolactone, especially in patients with a baseline potassium level above 4.5 mmol/L or GFR less than 45 mL/min/1.73 m². A more recent study in patients with advanced CKD (GFR 25 to $\leq$ 45 mL/min/1.73 m²) and resistant hypertension (the AMBER study) evaluated whether coadministration of patiromer (as potassium-binding polymer) with spironolactone could facilitate the safer use of spironolactone in these patients.⁴³ The study showed that hyperkalemia with spironolactone occurred much less commonly (1%) in patients coadministered patiromer versus spironolactone (7%), and BP was lowered significantly after the addition of spironolactone in these patients. The PATHWAY-2 study showed that amiloride in higher doses that were usually administered (10–20 mg daily) was as effective at spironolactone at lowering BP in resistant hypertension,¹⁹ but as with all potassium-sparing diuretics, similar caution is needed with respect to potassium levels. Other theoretical (but not proven) alternatives to spironolactone for treatment of resistant hypertension include eplerenone, higher doses of thiazide-type diuretics, loop diuretics in place of thiazide diuretics in patients with GFR less than 45

mL/min/1.73 m², the newer generation of nonsteroidal mineralocorticoid receptor antagonists, or even potentially newer drugs with multi-nephron segment diuretic actions such as ARNIs and SGLT2 inhibitors (see earlier).

Finally, treatments beyond further diuretic therapy may be effective for some patients with RH; for example, both doxazosin and bisoprolol did lower BP more effectively than placebo in the PATHWAY-2 study, though not as effectively as spironolactone. Alternatively, clonidine has also been shown to be effective at lowering BP in RH.⁴⁴ For patients where all else has failed, it may be necessary to resort to a combination of minoxidil, loop diuretic, and β -blocker to improve BP control.

Pharmacologic Treatment of Hypertension With Specific Comorbidities or in Specific Patient Groups

Hypertension With Diabetes, or Cerebrovascular Disease, Peripheral Vascular Disease, or Hypertension-Mediated Organ Damage

The core drug treatment algorithm in the European guidance (Fig. 37.3) is recommended for patients with uncomplicated hypertension, those with hypertension-mediated organ damage (e.g., LVH), or those with diabetes, cerebrovascular, or peripheral vascular disease.⁷ The only caveat is that the core algorithm may need to be modified in those with other comorbidities such as coronary artery disease (CAD), heart failure, or significant CKD (see later). The US guidance⁶ does not show a specific treatment algorithm but instead emphasizes which drug classes have compelling indications for specific clinical comorbidities—this generally leads to similar treatment as the European guidance for these conditions.

Hypertension in Patients With Coronary Artery Disease

All guidance acknowledges that in hypertensive patients with stable CAD, RAS blockers and β -blockers may improve outcomes post-MI, and that in patients with symptomatic angina, β -blockers and CCBs would be preferred components of the drug treatment strategy. Thus, combination therapy for patients with stable CAD differs little from the core treatment strategy, except for the addition of β -blockers as required.

Hypertension in Patients With Heart Failure

Heart failure with reduced ejection fraction (HFrEF) or preserved ejection fraction (HFpEF) are common consequences of long-standing

hypertension. Patients with HFrEF and hypertension should already be treated with guideline-directed therapy for their heart failure, including ACE inhibitors or ARBs, β -blockers, and mineralocorticoid receptor antagonists (e.g., spironolactone and eplerenone). If further BP lowering is required, thiazide/thiazide-like diuretics or DHP CCBs may be considered. ARNI therapy (sacubitril/valsartan) lowers BP (see earlier) and improves outcomes in patients with HFrEF, and it is indicated for the treatment of HFrEF as an alternative to ACE inhibitors or ARBs. In addition, SGLT2 inhibitors improve outcomes for patients with HFrEF (both diabetic and nondiabetic patients) and also more recently in HFpEF, and they may be considered for patients with heart failure to further reduce BP if required. Non-DHP CCBs (diltiazem and verapamil), α -blockers, and centrally acting agents, such as moxonidine, should not be used in patients with heart failure.

BP lowering is commonly required in patients with HFpEF. Effective treatments for HFpEF are less clearly defined, and in most patients, treatment will follow similar principles to the treatment of HFrEF. If volume overload is present, diuretics will be required.

Hypertension in Patients With Chronic Kidney Disease

There is consensus that RAS blockade should be part of drug treatment for hypertension in patients with CKD, especially when albuminuria or proteinuria are present.^{6,7,9} This would usually be prescribed with a CCB or diuretic, the latter specifically required when volume overload is evident, but required in most patients with more advanced CKD when three or more drugs may be required to control BP. Other key considerations are that (1) thiazide/thiazide-like diuretics often become less effective when eGFR is less than 45 mL/min/1.73 m² and certainly when less than 30 mL/min/1.73 m², and will need to be replaced with a loop diuretic if a diuretic is required; and (2) hyperkalemia becomes more of a concern, especially with potassium-sparing diuretics when eGFR is less than 45 mL/min/1.73 m² or baseline potassium levels are greater than 4.5 mmol/L.

Hypertension in Women

For a detailed discussion of this topic, the reader is referred to the US and European guidance,^{6,7} but it is important to highlight some specific considerations: (1) oral contraceptives (OCP), particularly those containing estrogen, may result in a significant rise in BP in about 5% of women, and thus BP should be carefully monitored and an alternative OCP formulation or form of contraception will usually be required in those who develop significant elevations of BP; (2) menopause is associated with an increased risk of hypertension, and the presence of hypertension is not a contraindication to hormone replacement therapy, provided that BP can be controlled; (3) in patients who are hypertensive and planning a pregnancy (i.e., prepregnancy hypertension), careful consideration should be given to whether BP medication is needed before consensus; when treatment is required, RAS blockers should be avoided, and the safest options are methyldopa, labetalol, or CCBs; there is controversy about whether diuretics should be used; and (4) postpregnancy, prescribers should refer to local formulary guidance regarding which drugs are safe to use when breastfeeding.

Hypertension in People of Black African Origin

Hypertension is more prevalent in Black patients, is associated with more target organ damage, and carries a worse prognosis, with a particularly high risk for stroke.⁴⁵ Black patients tend to respond better to diuretics, CCBs, and dietary salt restriction than White patients. ACE inhibitors, ARBs, and β -blockers are generally less effective as initial therapy in Black patients but become more effective in combination with diuretics or CCBs. When a RAS blocker is used as part of the treatment strategy, an ARB may be preferred because of the increased risk for angioedema with ACE inhibitors in Black patients.

Hypertension in Older People

If a BP of 140/90 mm Hg or higher is used to define hypertension, more than 70% of people older than 60 years will be hypertensive—most of these patients having isolated systolic hypertension (ISH). Surveys suggest that physicians consistently underestimate the risks and under-treat hypertension in older people. However, important considerations in treating older people include the following:

1. The arterial wall stiffening that gives rise to systolic hypertension and increased pulse pressure is also associated with impaired baroreflex sensitivity with increased risk for orthostatic hypotension. Thus, it is important to record lying and standing BP readings in elderly patients. However, contrary to popular perception, BP-lowering therapy has not been associated with increased risk of falls, even with lower BP goals now being advocated.
2. GFR declines with age, and older patients are more susceptible to reductions in GFR with BP lowering. In addition, they also often have impaired kidney conservation of sodium and fluid and may be more susceptible to volume depletion as a result of diuretic therapy.
3. Clearance of drugs and their active metabolites is decreased as a result of reduced hepatic and kidney function.
4. Cardiac function and reserve are often reduced, and patients are therefore much more likely to develop cardiac failure. Trials in elderly patients have consistently shown that treatment of hypertension reduces morbidity and mortality from cardiac failure.
5. Multiple comorbidities are much more common with advancing age, and this heterogeneity of patients means that actual age is a less important determinant of an individual patient's tolerability of BP lowering than their functional independence.
6. Communication and adherence with therapy may be more difficult with decline in cognitive function.

Despite these considerations, age should never be considered a contraindication to treatment, as most older patients will generally tolerate BP-lowering medications well, and BP reduction leads to substantial benefits, reducing the risk for stroke, coronary events, heart failure, and possibly dementia.⁴⁶ As a general rule, dosages should be increased more gradually and monotherapy initiation should be considered in older patients who are frail. Therapy needs to be tailored to individuals, remembering that elderly participants in clinical trials tend to be fitter and more independent than the many millions of elderly patients to whom guidelines are applied. However, it is reassuring for patients and families that the benefits are substantial if effective BP lowering can be achieved and is tolerated. Finally, there is a paucity of data with regard to treatment strategies for the most frail and dependent elderly patients, often with limited life expectancy, which in many cases is unlikely to be influenced by treatment of hypertension. Treatment decisions for such patients have to be individualized in consultation with the patient, family, and carers.

Concomitant Medication to Reduce Cardiovascular Risk

Treating hypertension should be considered part of a more comprehensive strategy to reduce CVD risk. Various tools to assess CVD risk, usually over a 10-year time frame, are available. The proposed level of CVD risk considered sufficient to warrant additional interventions has been progressively reduced toward 10% CVD risk over 10 years, and more recently, estimation of lifetime risk has been advocated.⁴⁷ Patients at high risk (i.e., with established CVD, organ damage, or diabetes or with a high calculated CVD risk) should be considered for additional interventions to reduce risk (i.e., reinforcement of lifestyle advice, especially smoking cessation). In addition, routine use of statins to reduce total cholesterol values by 40 mg/dL (~1 mmol/L) reduces the risk for coronary events by about one-third and for stroke by about one-fifth above the benefit already accrued from BP lowering.^{48,49} Moreover, this risk reduction associated

with statin therapy is not dependent on a high baseline cholesterol value. Hypertensive patients should be considered for treatment with antiplatelet drugs once BP has been controlled for secondary prevention (i.e., in those with established CVD or cerebrovascular disease).

Follow-up

In the early stages of treatment, the frequency of monitoring will be determined by the patient's response to therapy, the comorbidities, and the complexity of the treatment regimen required to control the BP. After initiating therapy, patients should be reviewed frequently to adjust treatment, monitor for adverse effects, and establish BP control. Once BP is controlled, patients should be reevaluated at least annually for a formal review, and most will be reevaluated every 6 months. Patients are increasingly monitoring their own BP between appointments, and this trend is likely to increase.

Withdrawal of Therapy

Most patients with hypertension require lifelong therapy. Some with milder hypertension who make major adjustments to their lifestyle may obtain sufficient fall in their BP to warrant safe withdrawal of monotherapy. However, patients with target organ damage or those at high CVD risk usually should not have their therapy withdrawn unless there is a compelling clinical reason.^{6,7} In patients with previously severe hypertension that has subsequently been well controlled, treatment withdrawal may not result in an immediate increase in BP, which can sometimes take many months to return to dangerously high values. Any patient who discontinues therapy must remain under review with regular BP monitoring. All but a very few will require treatment again.

Indications for Specialist Referral

Referral to a specialist center is sometimes indicated for the patient in management of hypertension. Indications include uncertainty about

BOX 37.1 Patient Characteristics That Should Raise Suspicion of Secondary Hypertension, and/or Circumstances That Should Prompt Consideration of Patient Referral to a Specialist Center for Further Evaluation

- Younger patients (<40 years) with grade 2 hypertension or onset of any grade of hypertension in childhood
- Acute worsening of hypertension in patients with previously documented chronically stable normotension
- Resistant hypertension (BP uncontrolled despite treatment with optimal best-tolerated doses of three or more drugs including a diuretic, and confirmed by ABPM or HBPM)
- Severe (grade 3) hypertension or a hypertension emergency
- Presence of extensive HMOD
- Clinical or biochemical features suggestive of endocrine causes of hypertension or CKD
- Clinical features suggestive of an OSA
- Symptoms suggestive of pheochromocytoma or family history of pheochromocytoma

Modified from Williams B, Mancia G, Spiering W, et al. 2018 ESC/ESH guidelines for the management of arterial hypertension. *Eur Heart J*. 2018;39(33):3021–3104. Erratum *Eur Heart J*. 2019;40(5):475. ABPM, Ambulatory blood pressure monitoring; BP, blood pressure; CKD, chronic kidney disease; HBPM, home blood pressure monitoring; HMOD, hypertension-mediated organ damage; OSA, obstructive sleep apnea.

the decision to treat, investigations to exclude secondary hypertension, severe and complicated hypertension, and resistant hypertension (Box 37.1).

SELF-ASSESSMENT QUESTIONS

- Which of the following statements is *true* about combination therapy for the treatment of hypertension?
 - Combination therapy with two drugs is recommended as initial therapy only for patients younger than 65 years.
 - There are three- or four-drug single-pill combinations now available, which are recommended as initial therapy for patients with hypertension.
 - The recommended combinations of drug therapy for most patients with hypertension usually involve a RAS blocker combined with a CCB or diuretic.
 - Single-pill combination therapy is associated with worse adherence with therapy than free combination of the same drugs.
 - Combinations of an angiotensin-converting enzyme inhibitor and ARB are recommended for some specific patient groups with hypertension.
- Which of the following statements is *true* of recommended BP targets for treated hypertensive patients?
 - Lowering BP to at most 140/90 mm Hg in all treated patients, if tolerated, is recommended by all guidelines.
 - The US guideline recommends routinely lowering BP to less than 120/80 mm Hg in high-risk patients.
 - The evidence for optimal BP targets is stronger for diastolic versus systolic BP.
 - All guidelines recommend the routine use of ambulatory or home BP monitoring for checking BP control in all patients.
 - The BP targets are similar for office, home, and ambulatory BP measurements.
- Which of the following statements is *true* of drugs used to treat hypertension?
 - Thiazide diuretics are more effective than thiazide-like diuretics at lowering BP.
 - RAS-blocking drugs are suitable treatment for women of child-bearing potential, either alone or as part of combination treatment, as long as they are discontinued by the third trimester in pregnant women.
 - β -Blockers are only used for specific guideline-directed indications and are especially effective as part of combination therapy in reducing the risk for stroke in older patients.
 - CCBs are less likely to reduce BP variability compared with other drug classes, especially in older patients.
 - Angiotensin receptor-neprilysin inhibitors are significantly more effective at lowering BP than ARBs alone.
- Which of the following statements is *true* for RH?
 - RH is common, affecting approximately 30% of treated patients.
 - When patients are thoroughly investigated, RH is usually found to be due to a secondary cause of hypertension.
 - RH is more common in younger patients.
 - It is only rarely possible to control BP in RH with additional drug therapy.
 - RH is most effectively treated with further diuretic therapy.

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